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EDUCATION

University of Massachusetts Dartmouth

Master of Science in Computer and Information Science

Jan' 2023 - Aug' 2024

Rajalakshmi Engineering College, Tamil Nadu, India

Bachelor of Engineering in Computer Science

Aug' 2018 - May 2022

ACADEMIC PROJECTS

Real-time Serverless Chat Application

Aug' 2024 - Sep' 2024

University of Massachusetts Dartmouth

- Developed a real-time chat application using **React.js**, **CSS3**, and **JavaScript**, integrating **Firebase** for authentication, real-time data synchronization, and cloud storage, ensuring a responsive **user experience** with **<100ms latency**.
- Integrated **Firebase Authentication** for secure user login and registration, incorporating encrypted data storage to protect user credentials and ensure a high security compliance rate.
- Built an interactive **user interface** with **CSS3** and **React Hooks**, applying flexible design principles to optimize usability across devices, resulting in a **35%** increase in **user engagement** and **30%** reduction in session duration.
- Leveraged **Zustand** for **state management**, optimizing asynchronous operations and state handling to ensure an uninterrupted, lag-free chat experience with real-time updates.
- Incorporated **WebRTC** for peer-to-peer communication, allowing real-time audio and video streaming between users without the need for additional servers, enhancing interaction features.

Decentralized NFT Marketplace and Token Transfer using ICP Blockchain

Jun' 2024 - Jul' 2024

University of Massachusetts Dartmouth

- Engineered a **decentralized token system** allowing users to authenticate via **ICP Internet Identity**, claim tokens through a faucet, perform token transfers, and maintain a detailed transaction history, enhancing platform security and user trust.
- Orchestrated an NFT marketplace where users can mint, buy, and sell NFTs, leveraging inter-canister communication to synchronize NFT transactions with the token transfer system, ensuring 100% transparent and consistent transaction tracking.
- Implemented **Motoko-based smart contracts** to facilitate automated and secure transactions within the marketplace, reducing transaction errors and improving processing efficiency by **45%**.
- Improved frontend architecture using **React.js (React Router)**, **HTML**, **JavaScript** and **CSS3** for a smooth user interface, improving engagement and interaction rates by **35%**.
- Ensured **system scalability** to handle an increasing volume of transactions and user activity, achieving **70%** higher throughput without performance degradation than other blockchains, backed by rigorous **Git version control** for seamless development and deployment.

To-Do List Application

May 2024 - Jun' 2024

University of Massachusetts Dartmouth

- Created a **To-Do List** web application with **JavaScript**, **Embedded JavaScript (EJS)**, **Express.js**, **HTML**, and **CSS**, ensuring robust integration between frontend and backend components for a seamless experience.
- Enhanced **data management** by implementing **PostgreSQL**, achieving a **40%** increase in scalability and performance through efficient data storage and retrieval processes.
- Designed and deployed **RESTful APIs** to enable smooth and secure communication between the **client-side** and **server-side** systems, facilitating real-time data interactions and speeding up system responsiveness.
- Utilized **Material-UI** and responsive web design principles using **React.js** to craft a visually appealing interface, resulting in a **30%** improvement in overall user experience.

SKILLS

- **Frontend:** HTML5, CSS3, Bootstrap 5, JavaScript (ES6+), React.js, Zustand, Redux (state management), jQuery.
- **Backend:** Node.js, Express.js, EJS, RESTful APIs, GraphQL, Motoko (smart contracts), Cloud Firestore, NPM, WebRTC.
- **Databases:** PostgreSQL, MongoDB (NoSQL), MySQL (RDBMS), Firebase Database.
- **Programming Languages:** Python (Django), Java, C, C++, PHP.
- **Tools:** Git and GitHub (version control), Command Line Interface (CLI), Visual Studio Code, Microsoft Office Suite, AWS.
- **Additional Skills:** Verbal Communication, Teamwork, Software Development Lifecycle (SDLC).